

AHMED ELBAKLY

Senior Full-Stack Engineer | Node.js • NestJS • React • Next.js • TypeScript • Cloud & DevOps

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 5+ years shipping production SaaS platforms end to end on Node.js, NestJS, React, Next.js and TypeScript. Architected multi-tenant backends of 90+ feature modules and enterprise frontends exceeding 144 pages, combining secure REST APIs, JWT and RBAC authorization, Redis caching and Socket.IO real-time features with performant, accessible React interfaces. Owns the full delivery cycle — system design, database modeling, state architecture, payments, i18n/RTL, Docker, GitLab CI/CD and cloud deployment on AWS, Azure AKS and DigitalOcean. Sole technical owner on 9+ client platforms across real estate, EdTech, e-commerce and on-demand services. Bilingual (Arabic/English) engineer serving MENA, European and remote US teams.

CORE COMPETENCIES

Full-Stack Architecture • System Design • REST API Design • Multi-Tenant SaaS Platforms • Microservices Patterns • Component & State Architecture • Authentication & Authorization • JWT • RBAC • API Security • Redis Caching • Performance Optimization • Real-Time Systems (WebSockets) • Event-Driven & Queue-Based Processing • Payments & Subscription Billing • Internationalization & RTL • Cloud Deployment • CI/CD Automation • Containerization & Orchestration • Technical Ownership & Client Delivery

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Frontend: React 18/19, Next.js (App Router), Redux Toolkit, RTK Query, TanStack Query, Material UI, Tailwind CSS, CASL, Chart.js, Vite, i18n & RTL (EN-AR)

Backend: Node.js, NestJS, Express.js, REST APIs, GraphQL, Microservices Patterns, Multi-Tenant Architecture, Socket.IO / WebSockets, BullMQ, Cron Jobs, JWT, RBAC

Databases & Caching: MongoDB (Mongoose), PostgreSQL, MySQL, Redis, Prisma

Cloud: AWS (EC2, S3), Azure AKS, DigitalOcean, Cloudflare (DNS / SSL / CDN)

DevOps: Docker, Kubernetes, GitLab CI/CD, Nginx, Linux (Ubuntu)

Testing: Jest, Postman

Integrations: Stripe (billing & subscriptions), Keycloak (OIDC / OAuth2), Firebase, Twilio, Agora, LiveKit, Google Maps, Mapbox

Tools & Practices: Git, GitHub, GitLab, Agile / Scrum, Code Review

PROFESSIONAL EXPERIENCE

Senior Full-Stack Engineer — Freelance / Contract May 2020 – Present

Remote — SaaS, CRM, EdTech, E-Commerce & On-Demand Platforms

- Designed and shipped 9+ production web platforms end to end for clients in real estate, education, e-commerce and on-demand services, owning everything from data model to deployed UI.
- Architected multi-tenant backends in NestJS and Node.js with JWT authentication, role-based access control, audit logging and BullMQ background processing, so each client could scale on a shared, secure codebase.
- Engineered enterprise React 18/19 and Next.js frontends — including a 144+ page CRM and multi-role dashboards for six user types — using Redux Toolkit, RTK Query and TanStack Query to keep large interfaces predictable and maintainable.
- Delivered real-time chat, notifications and live order tracking over Socket.IO, replacing polling with event-driven updates that kept operational dashboards current.
- Integrated Stripe billing and subscriptions, Keycloak identity, Firebase, Twilio SMS, Agora and LiveKit streaming, Google Maps and Mapbox, expanding product capability without rebuilding core services.
- Optimized performance across the stack with Redis caching, tuned MongoDB and PostgreSQL queries and Next.js App Router rendering and image optimization, improving response and load times.
- Automated delivery by containerizing applications with Docker, building GitLab CI/CD pipelines and deploying to AWS, Azure AKS (Kubernetes) and DigitalOcean behind Nginx and Cloudflare DNS, SSL and CDN.
- Localized products for bilingual English/Arabic audiences with full RTL support and internationalization, extending client reach across MENA markets.
- Led delivery end to end as the sole technical owner — requirements, architecture, estimation, code review and production support — for clients without an in-house engineering team.

KEY PROJECTS

Coonex — Real Estate CRM SaaS (Platform + Admin Dashboard)

Full-Stack Engineer

Next.js 13 · React 18 · TypeScript · Node.js · Express.js · MongoDB · Redux Toolkit · MUI · CASL · Socket.IO · Stripe

- Replaced spreadsheet-driven agency workflows with a multi-tenant CRM managing 500+ properties and 1,000+ leads across lead management, campaigns, sales pipelines, meetings, documents and analytics.
- Engineered an enterprise frontend of 144+ pages and 12+ business modules with CASL-based permissions and a disciplined state architecture, keeping a large surface area maintainable as features grew.
- Monetized the platform through Stripe billing and subscription management, and surfaced instant updates via Socket.IO notifications.
- Shipped a separate B2B admin dashboard with RBAC, analytics, reporting and invoice management, giving the operator central control over tenants and revenue.

Schoolie — Multi-Tenant School Management SaaS Full-Stack Engineer

NestJS · React 18 · TypeScript · Redux Toolkit · RTK Query · MongoDB · AWS S3 · Firebase · Twilio · Tailwind CSS · Jest

- Unified fragmented school administration into one SaaS platform with 43 backend modules spanning academics, attendance, payroll, fee collection, communication and reporting.
- Developed a 25+ module multi-role frontend for Admin, School Manager, Teacher, Student, Parent and Driver users, with granular RBAC and permission management restricting sensitive data per role.
- Delivered real-time chat, push notifications, maps, analytics dashboards and complete English/Arabic support, widening adoption across bilingual MENA schools.
- Automated recurring operations with cron jobs, integrated Twilio, Firebase, AWS S3 and Excel export pipelines, and covered core services with Jest tests.

49Hub — Multi-Service Super-App Backend Engineer

Node.js · TypeScript · Express.js · MongoDB · Redis · Socket.IO · BullMQ · AWS S3

- Architected backend services powering 90+ feature modules across ride-hailing, food delivery, hotel booking, live streaming, social networking and marketplace verticals in a single platform.
- Reduced database load and improved response times with Redis caching, and offloaded heavy work to BullMQ workers and cron jobs to protect API latency.
- Hardened access with RBAC and audit logging, and integrated Agora, LiveKit, Google Maps, Mapbox, Firebase, AWS S3 and multiple payment gateways.

BaladiCRM — CRM Platform with AI Chatbot Builder Frontend Engineer

React 18 · Redux Toolkit · Socket.IO · Material UI · Tailwind CSS · Vite

- Engineered a CRM with 10+ business modules, real-time communication and a notification system, giving teams a single operational workspace.
- Delivered an AI-powered chatbot flow builder with training and approval workflows, letting non-technical staff configure automated conversations safely.
- Designed a scalable Redux architecture of 21+ slices with JWT authentication and RBAC, keeping state predictable as the module count grew.

Jumla & Luminar — E-Commerce Platforms Frontend Engineer

React 19 · Next.js · TypeScript · Redux Toolkit · TanStack Query · Material UI · Tailwind CSS · Docker · Kubernetes · Azure AKS

- **Jumla** — Enhanced a production multi-vendor e-commerce and delivery dashboard with real-time order tracking, RBAC and Google Maps geolocation; participated in deployment and maintenance on Docker, Kubernetes and Azure AKS.
- **Luminar** — Built a feature-based admin dashboard covering products, orders, payments, revenue reporting, analytics and audit logs, plus a multilingual EN/AR corporate site with full RTL/LTR support optimized via the Next.js App Router.

CLOUD & DEVOPS EXPERIENCE

- Deployed and maintained production applications on AWS, Azure AKS, DigitalOcean and Hostinger, covering server provisioning, monitoring and live incident troubleshooting.
- Containerized applications with Docker and built GitLab CI/CD pipelines for automated testing, deployment and release.
- Configured Nginx reverse proxies and managed domains, DNS, SSL and CDN through Cloudflare to improve availability and transport security.

EDUCATION & TRAINING

- **Bachelor of Commerce** — Nile Academy for Science, Egypt (2012)
- **Web Development — Advanced, Professional & Challenger Tracks** — Udacity (FWD Initiative)

LANGUAGES

Arabic — Native | English — Professional Working Proficiency